Chapter 4

Minimal flows, divergence constraints and transport density

4.1 Eulerian and Lagrangian points of view

We review here the main languages and the main mathematical tools to describe static or dynamical transport phenomena.

4.1.1 Static and Dynamical models

This section presents a very informal introduction to the physical interpretation of dynamical models in optimal transport.

In fluid mechanics, and in many other topics with similar modeling issues, it is classical to consider two complementary ways of describing motions, which are called Lagrangian and Eulerian.

When we describe a motion via Lagrangian formalism we give "names" to particles (using either a specific label, or their initial position, for instance) and then describe, for every time t and every label, what happens to that particle. "What happens" means providing its position and/or its speed. Hence we could for instance look at trajectories $y_x(t)$, standing for the position at time t of particle originally located at x. As another possibility, instead of giving names we could consider bundles of particles with the same behavior and indicate how many are they. This amounts to giving a measure on possible behaviors.

The description may be more or less refined. For instance if one only considers two different times $t = 0$ and $t = 1$, the behavior of a particle is only given by its initial and final positions. A measure on those pairs (x, y) is exactly a transport plan. This explains why we can consider that the Kantorovich problem is expressed in Lagrangian coordinates. The Monge problem is also Lagrangian, and particles are labelled by their initial position.

More refined models can be easily conceived: reducing a movement to the initial and final positions is embarrassingly poor! Measures on the set of paths (curves $\omega : [0,1] \to \Omega$, with possible assumptions on their regularity) have been used in many modeling, and in particular in traffic issues, branched transport (see the Section 4.4 for both these subjects), or in Brenier's variational formulation of the incompressible Euler equations for fluids (see Section 1.7.4, 8.4.4 and [84, 86, 53]).

On the other hand, in the Eulerian formalism, we describe, for every time t and every point x, what happens at such a point at such a time. "What happens" usually means what are the velocity, the density and/or the flow rate (both in intensity and in direction) of particles located at time t at point x.

Eulerian models may be distinguished into static and dynamical ones. In a dynamical model we usually use two variables, i.e. the density $\rho(t,x)$ and the velocity $\mathbf{v}_t(x)$. It is possible to write the equation satisfied by the density of a family of particles moving according to the velocity field $\mathbf{v}$. This means that we prescribe the initial density ρ_0, and that the position of the particle originally located at x will be given by the solution of the ODE

$$\begin{cases} y_x'(t) = \mathbf{v}_t(y_x(t)) \\ y_x(0) = x. \end{cases} \tag{4.1}$$

We define the map Y_t through $Y_t(x) = y_x(t)$, and we look for the measure $\rho_t := (Y_t)_\#\rho_0$. It is well known that ρ_t and $\mathbf{v}_t$ solve together the so-called continuity equation

$$\partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}_t) = 0$$

that is briefly addressed in Section 4.1.2.

The static framework is a bit harder to understand, since it is maybe not clear what "static" means when we want to describe movement. One has to think to a permanent, cyclical movement, where some mass is constantly injected into the motion at some points and constantly withdrawn somewhere else. We can also think at a time average of some dynamical model: suppose that we observe the traffic in a city and we wonder what happens at each point, but we do not want an answer depending on the hour of the day. We could for instance consider as a traffic intensity at every point the average traffic intensity at such a point on the whole day. In this case we usually use a unique variable $\mathbf{w}$ standing for the mass flow rate (which equals density times speed: hence, it is rather a momentum than a velocity), and the divergence $\nabla\cdot\mathbf{w}$ stands for the excess of mass which is injected into the motion at every point. More precisely, if particles are injected into the motion according to a density μ and then exit with density ν, the vector fields $\mathbf{w}$ standing for flows connecting these two measures must satisfy

$$\nabla\cdot\mathbf{w} = \mu - \nu.$$

4.1.2 The continuity equation

This section is devoted to the equation

$$\partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}_t) = 0,$$

its meaning, formulations, and uniqueness results. The remainder of the chapter will mainly deal with the static divergence equation, but we will see later that the dynamical case can also be useful (in particular, to produce transport plans associated to a given a vector field). Hence, we need to develop some tools.

First, let us spend some words on the notion of solution for this equation. Here below $\Omega \subset \mathbb{R}^d$ is a bounded domain, or $\mathbb{R}^d$ itself.

Definition 4.1. We say that a family of pairs measures/vector fields $(\rho_t, \mathbf{v}_t)$ with $\mathbf{v}_t \in L^1(\rho_t; \mathbb{R}^d)$ and $\int_0^T ||\mathbf{v}_t||_{L^1(\rho_t)} \mathrm{d}t = \int_0^T \int_\Omega |\mathbf{v}_t| \mathrm{d}\rho_t \, \mathrm{d}t < +\infty$ solves the continuity equation on $]0,T[$ in the distributional sense if for any bounded and Lipschitz test function $\phi \in C^1_c(]0,T[\times\overline{\Omega})$, (note that we require the support to be far from $t = 0,1$ but not from $\partial\Omega$, when ω is bounded; also Ω is usually supposed to be itself closed but we write $\overline{\Omega}$ to stress the fact that we do include its boundary), we have

$$\int_0^T \int_\Omega (\partial_t \phi) \, \mathrm{d}\rho_t \, \mathrm{d}t + \int_0^T \int_\Omega \nabla\phi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t \, \mathrm{d}t = 0. \tag{4.2}$$

This formulation includes homogeneous Neumann boundary conditions on $\partial\Omega$ for $\mathbf{v}_t$ (if Ω is not $\mathbb{R}^d$ itself, obviously), i.e. $\rho_t \mathbf{v}_t \cdot \mathbf{n} = 0$. If we want to impose the initial and final measures we can say that $(\rho_t, \mathbf{v}_t)$ solves the same equation, in the sense of distribution, with initial and final data ρ_0 and ρ_T, respectively, if for any test function $\phi \in C^1_c([0,T]\times\overline{\Omega})$ (now we do not require the support to be far from $t = 0,1$), we have

$$\int_0^T \int_\Omega (\partial_t \phi) \, \mathrm{d}\rho_t \, \mathrm{d}t + \int_0^T \int_\Omega \nabla\phi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t \, \mathrm{d}t = \int_\Omega \phi(T,x) \, \mathrm{d}\rho_T(x) - \int_\Omega \phi(0,x) \, \mathrm{d}\rho_0(x). \tag{4.3}$$

We can also define a weak solution of the continuity equation through the following condition: we say that $(\rho_t, \mathbf{v}_t)$ solves the continuity equation in the weak sense if for any test function $\psi \in C^1_c(\overline{\Omega})$, the function $t \mapsto \int \psi \, \mathrm{d}\rho_t$ is absolutely continuous in t and, for a.e. t, we have

$$\frac{d}{dt} \int_\Omega \psi \, \mathrm{d}\rho_t = \int_\Omega \nabla\psi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t.$$

Note that in this case $t \mapsto \rho_t$ is automatically continuous for the weak convergence, and imposing the values of ρ_0 and ρ_1 may be done pointwisely in time.

Proposition 4.2. *The two notions of solutions are equivalent: every weak solution is actually a distributional solution and every distributional solution admits a representative (another family $\widetilde{\rho}_t = \rho_t$ for a.e. t) which is weakly continuous and is a weak solution.*

Proof. To prove the equivalence, take a distributional solution, and test it against functions ϕ of the form $\phi(t,x) = a(t)\psi(x)$, with $\psi \in C^1_c(\overline{\Omega})$ and $a \in C^1_c(]0,1[)$. We get

$$\int_0^T a'(t) \int_\Omega \psi(x) \, \mathrm{d}\rho_t \, \mathrm{d}t + \int_0^1 a(t) \int_\Omega \nabla\psi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t \, \mathrm{d}t = 0.$$

The arbitrariness of a shows that the distributional derivative (in time) of $\int_\Omega \psi(x) \, \mathrm{d}\rho_t$ is $\int_\Omega \nabla\psi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t$. This last function is L^1 in time since $\int_0^T \left| \int_\Omega \nabla\psi \cdot \mathbf{v}_t \, \mathrm{d}\rho_t \right| \mathrm{d}t \le \mathrm{Lip}\,\psi \int_0^T ||\mathbf{v}_t||_{L^1(\rho_t)} \mathrm{d}t < +\infty$. This implies that $(\rho, \mathbf{v})$ is a weak solution.

Conversely, the same computations shows that weak solutions satisfy (4.2) for any ϕ of the form $\phi(t,x) = a(t)\psi(x)$. It is then enough to prove that finite linear combination of these functions are dense in $C^1([0,T]\times K)$ for every compact set $K \subset \mathbb{R}^d$ (this is true, but is a non-trivial exercise: **Ex**(23)). $\square$ $\square$

It is also classical that smooth functions satisfy the equation in the classical sense if and only if they are weak (or distributional) solutions. We can also check the following.

Proposition 4.3. *Suppose that ρ is Lipschitz continuous in (t,x), that $\mathbf{v}$ is Lipschitz in x, and that the continuity equation $\partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0$ is satisfied in the weak sense. Then the equation is also satisfied in the a.e. sense.*

Proof. First we note that with our assumptions both $\partial_t\rho$ and $\nabla\cdot(\rho\mathbf{v})$ are well defined a.e. Fix a countable set $D \subset C^1_c(\mathring{\Omega})$ which is dense for the uniform convergence in $C^0_c(\mathring{\Omega})$ (for instance, use polynomial functions with rational coefficients multiplied times suitable cutoff functions). Fix t_0 such that $(t,x)\mapsto\rho(t,x)$ is differentiable at (t_0,x) for a.e. x, and also such that $t\mapsto\int_\Omega\phi\,\mathrm{d}\rho_t$ is differentiable at $t = t_0$ with derivative given by $\int_\Omega\nabla\phi\cdot\mathbf{v}_t\,\mathrm{d}\rho_t$ for all $\phi\in D$. Almost all t_0 satisfy these conditions. Then we can also write, by differentiating under the integral sign,

$$\frac{d}{dt}_{|t=t_0}\int_\Omega\phi\,\mathrm{d}\rho_t\,\mathrm{d}x = \int_\Omega\phi\,(\partial_t\rho)_{t_0}\,\mathrm{d}x,$$

which proves $\int_\Omega(\nabla\phi\cdot\mathbf{v}_{t_0})\rho_{t_0}\,\mathrm{d}x = \int\phi\,(\partial_t\rho)_{t_0}\,\mathrm{d}x$. Yet, we can write $\int_\Omega\nabla\phi\cdot\mathbf{v}_{t_0}\rho_{t_0}\,\mathrm{d}x$ as $-\int_\Omega\phi\,\nabla\cdot(\mathbf{v}_{t_0}\rho_{t_0})\,\mathrm{d}x$ (by integration-by-parts of Lipschitz functions, with no boundary term because of the compact support of ϕ), and finally we get

$$\int_\Omega\phi\left((\partial_t\rho)_{t_0}+\nabla\cdot(\rho_{t_0}\mathbf{v}_{t_0})\right)\mathrm{d}x = 0 \quad \text{for all } \phi\in D,$$

which is enough to prove $(\partial_t\rho)_{t_0}+\nabla\cdot(\rho_{t_0}\mathbf{v}_{t_0}) = 0$ a.e. in x. $\square$ $\square$

After checking the relations between these different notions of solutions, form now on we will often say “solution” to mean, indifferently, “weak solution” or “solution in the distributional sense”.

We now try to identify the solutions of the continuity equation and we want to connect them to the flow of the vector field $\mathbf{v}_t$. First, we recall some properties of the flow.

Box 4.1. **– Memo** – Flow of a time-dependent vector field

We consider the ODE $y'_x(t) = \mathbf{v}_t(y_x(t))$ with initial datum $y_x(0) = x$, as in (4.1).

Proposition - If $\mathbf{v}_t$ is continuous in x for every t, then for every initial datum there is local existence (there exists at least a solution, defined on a neighborhood of $t = 0$). If moreover $\mathbf{v}$ satisfies $|\mathbf{v}_t(x)| \le C_0 + C_1|x|$, then the solution is global in time. If $\mathbf{v}_t$ is Lipschitz in x, uniformly in t, then the solution is unique, and defines a flow $Y_t(x) := y_x(t)$. In this case, if we set $L := \sup_t \mathrm{Lip}(\mathbf{v}_t)$, then the map $Y_t(\cdot)$ is Lipschitz in x, with constant

$e^{L|t|}$. Moreover, $Y_t : \mathbb{R}^d \to \mathbb{R}^d$ is invertible and its inverse is also Lipschitz with the same constant.

Sketch of proof - We do not prove existence, which can be classically obtained by fixed-point arguments in the space of curves. As for uniqueness, and for the dependence on the initial datum x, we consider two curves $y^{(1)}$ and $y^{(2)}$, solutions of $y'(t) = \mathbf{v}_t(y(t))$, and we define $E(t) = |y^{(1)}(t) - y^{(2)}(t)|^2$. We have

$$E'(t) = 2(y^{(1)}(t) - y^{(2)}(t)) \cdot (\mathbf{v}_t(y^{(1)}(t)) - \mathbf{v}_t(y^{(2)}(t))),$$

which proves $|E'(t)| \leq 2LE(t)$. By Gronwall's Lemma, this gives $E(0)e^{-2L|t|} \leq E(t) \leq E(0)e^{2L|t|}$ and provides at the same time uniqueness, injectivity, and the bi-Lipschitz behavior of Y_t (which is also a homeomorphism from $\mathbb{R}^d$ onto $\mathbb{R}^d$ since for every $x_0 \in \mathbb{R}^d$ we can solve the ODE imposing the Cauchy datum $y(t) = x_0$).

Then, we can prove the following, that we state for simplicity in the case of Lipschitz and bounded vector fields.

Theorem 4.4. *Suppose that $\Omega \subset \mathbb{R}^d$ is either a bounded domain or $\mathbb{R}^d$ itself. Suppose that $\mathbf{v}_t : \Omega \to \mathbb{R}^d$ is Lipschitz continuous in x, uniformly in t, and uniformly bounded, and consider its flow Y_t. Suppose that, for every $x \in \Omega$ and every $t \in [0,T]$, we have $Y_t(x) \in \Omega$ (which is obvious for $\Omega = \mathbb{R}^d$, and requires suitable Neumann conditions on $\mathbf{v}_t$ otherwise). Then, for every probability $\rho_0 \in \mathscr{P}(\Omega)$ the measures $\rho_t := (Y_t)_\# \rho_0$ solve the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$ with initial datum ρ_0. Moreover, every solution of the same equation with $\rho_t \ll \mathscr{L}^d$ for every t is necessarily obtained as $\rho_t = (Y_t)_\# \rho_0$. In particular, the continuity equation admits a unique solution.*

Proof. First we check the validity of the equation when ρ_t is obtained from such a flow through (4.1). We will prove that we have a weak solution. Fix a test function $\phi \in C^1(\mathbb{R}^d)$ such that both ϕ and $\nabla\phi$ are bounded, and compute

$$\begin{aligned}\frac{d}{dt}\int_\Omega \phi \,\mathrm{d}\rho_t = \frac{d}{dt}\int \phi(y_x(t))\,\mathrm{d}\rho_0(x) &= \int_\Omega \nabla\phi(y_x(t)) \cdot y_x'(t)\,\mathrm{d}\rho_0(x) \\ &= \int_\Omega \nabla\phi(y_x(t)) \cdot \mathbf{v}_t(y_x(t))\,\mathrm{d}\rho_0(x) = \int_\Omega \nabla\phi(y) \cdot \mathbf{v}_t(y)\,\mathrm{d}\rho_t(y).\end{aligned}$$

This characterizes weak solutions.

In order to prove the second part of the statement, we first observe that (4.2), i.e.

$$\int_0^T \int_\Omega (\partial_t \phi + \mathbf{v}_t \cdot \nabla\phi)\,\mathrm{d}\rho_t\,\mathrm{d}t = 0,$$

is also valid for all Lipschitz compactly supported test functions ϕ, whenever $\rho_t \ll \mathscr{L}^d$ for every t. Indeed, if we fix a Lipschitz test function ϕ and we smooth it by convolution with a compactly supported kernel, we have a sequence $\phi_\varepsilon \in C_c^\infty$ such that $\nabla_{t,x}\phi_\varepsilon \to \nabla_{t,x}\phi$ a.e. and we can apply dominated convergence since $|\nabla_{t,x}\phi_\varepsilon| \leq \mathrm{Lip}(\phi)$ (we need ρ_t to be absolutely continuous because we only have Lebesgue-a.e. convergence).

Then we take a test function $\psi \in C^1_c(\mathbb{R}^d)$ and we define $\phi(t,x) := \psi((Y_t)^{-1}(x))$. If we can prove that $t \mapsto \int \phi(t,x)\,\mathrm{d}\rho_t(x)$ is constant, then we have proven $\rho_t = (Y_t)_\#\rho_0$. The function ϕ is Lipschitz continuous, because the flow Y_t is bi-Lipschitz; yet, it is not compactly supported in time (it is compactly supported in space since $\operatorname{spt}\phi$ is compact and, if we set $M := \sup_{t,x}|\mathbf{v}_t(x)|$, we see that $\phi(t,x)$ vanishes on all points x which are at distance larger than tM from $\operatorname{spt}\psi$). Thus, we multiply it times a cut-off function $\chi(t)$, with $\chi \in C^1_c(]0,T[)$. We have

$$\partial_t(\chi\phi) + \mathbf{v}_t \cdot \nabla(\chi\phi) = \chi'(t)\phi(t,x) + \chi(t)\left(\partial_t\phi(t,x) + \mathbf{v}_t(x)\cdot\nabla\phi(t,x)\right).$$

We can prove that, by definition of ϕ, the term $\partial_t\phi(t,x) + \mathbf{v}_t(x)\cdot\nabla\phi(t,x)$ vanishes a.e. (this corresponds to saying that ϕ is a solution of the transport equation, see Box 6.3). This is true since we have $\phi(t,T_t(x)) = \psi(x)$ and, differentiating it w.r.t. t (which is possible for a.e. (t,x)), we get $\partial_t(t,Y_t(x)) + \mathbf{v}_t(Y_t(x))\cdot\nabla\phi(t,Y_t(x)) = 0$, which means that $\partial_t\phi + \mathbf{v}_t\cdot\nabla\phi$ vanishes everywhere, as Y_t is surjective.

Hence, from the definition of distributional solution we have

$$\int_0^T \chi'(t)\mathrm{d}t \int_{\mathbb{R}^d} \phi(t,x)\,\mathrm{d}\rho_t(x) = \int_0^T \mathrm{d}t \int_{\mathbb{R}^d} \left(\partial_t(\chi\phi) + \mathbf{v}_t\cdot\nabla(\chi\phi)\right)\mathrm{d}\rho_t(x) = 0.$$

The test function $\chi \in C^1_c(]0,T[)$ being arbitrary, we get that $t \mapsto \int_{\mathbb{R}^d}\phi(t,x)\,\mathrm{d}\rho_t(x)$ is constant. □ □

4.2 Beckmann's problem

In this section we discuss a flow-minimization problem introduced by Beckmann in the '50s as a particular case of a wider class of convex optimization problem, of the form $\min\{\int H(\mathbf{w})\mathrm{d}x \,:\, \nabla\cdot\mathbf{w} = \mu - \nu\}$, for convex H. In this section we discuss the case $H(z) = |z|$, which is a limit case in the class of costs studied by Beckmann, who was indeed more interested in strictly convex costs. Strict convexity will be considered in the Discussion section 4.4.1. The reason to stick to the linear cost here, and to spend more words on it than on the general case, lies in its equivalence with the Monge Problem. We will provide all the tools to study this equivalence and this problem.

4.2.1 Introduction, formal equivalences and variants

We consider here a simple version of a problem that has been proposed by Beckmann as a model for optimal transport in the '50s. Beckmann called it *continuous transportation model* and he was not aware of Kantorovich's works and the possible links between the two theories.

Beckmann's minimal flow problem

Problem 4.5. Consider the minimization problem

$$\text{(BP)} \quad \min\left\{\int |\mathbf{w}(x)|\,\mathrm{d}x \,:\, \mathbf{w}:\Omega\to\mathbb{R}^d,\, \nabla\cdot\mathbf{w} = \mu - \nu\right\}, \tag{4.4}$$